Quiz — 2.1.1 Thinking Abstractly — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Answer key

Section A (6 marks — 1 each) 1. **B** — modelling one specific thing (the road network) as a simpler representation (a graph) is representational abstraction. 2. **C** — grouping savings and current accounts by their shared characteristics into one general type is generalisation/categorisation. 3. **C** — the physical material/weight has no bearing on play; the game only needs piece positions, legal moves and whose turn it is. 4. **B** — breaking into sub-problems is decomposition; abstraction *removes/hides* detail. Common trap. 5. **B** — discarding true geography leaves only what matters for choosing a route (lines, order of stations, interchanges). 6. **B** — representing continuous reality as a simplified discrete grid is abstraction (building a workable model).

Section B (9 marks)

7. **[3]** Keep (any two, 1 each): position/coordinates on track; speed/velocity; direction/heading; fuel or tyre state if it affects play. Discard + justify (1): e.g. engine chemistry / exact paint colour / VIN — these do not affect how the car behaves or is driven in the race, so they are irrelevant to the simulation. Must justify the discard for the mark.

8. **[2]** Representational abstraction = simplifying *one specific* thing into a model (1) — e.g. a tube map of one city's network. Generalisation = grouping *many* things by shared characteristics into a reusable model (1) — e.g. one `Vehicle` class for cars and lorries. Award only with a valid example for each idea.

9. **[2]** Nodes represent junctions/places and edge weights represent the cost of travelling between them (distance or time) (1); this is appropriate because a shortest-path algorithm can then work purely on the weights, ignoring irrelevant detail like scenery or surface, to find the fastest route (1).

10. **[2]** A general model captures behaviour common to all cases (1), so a new case (e.g. a triangle) can be added by reusing/extending the existing model rather than rewriting code, reducing effort and bugs (1).

Section C (5 marks) 11. **[5]** Up to 5 from: Abstraction means building a simplified model that keeps only detail relevant to managing appointments (1). Keep — patient ID/name, date of birth, contact number, GP/clinic, required appointment type/duration (any, 1). Discard — e.g. eye colour, favourite food, home décor (1). Justification — such detail plays no part in scheduling or contacting the patient, so including it would add complexity without value (1). Reasoned link that removing irrelevant detail makes the system simpler, faster to build and easier to maintain while still solving the appointments problem (1). Must apply to *this* scenario, not give a generic definition.

Total: 6 + 9 + 5 = 20